

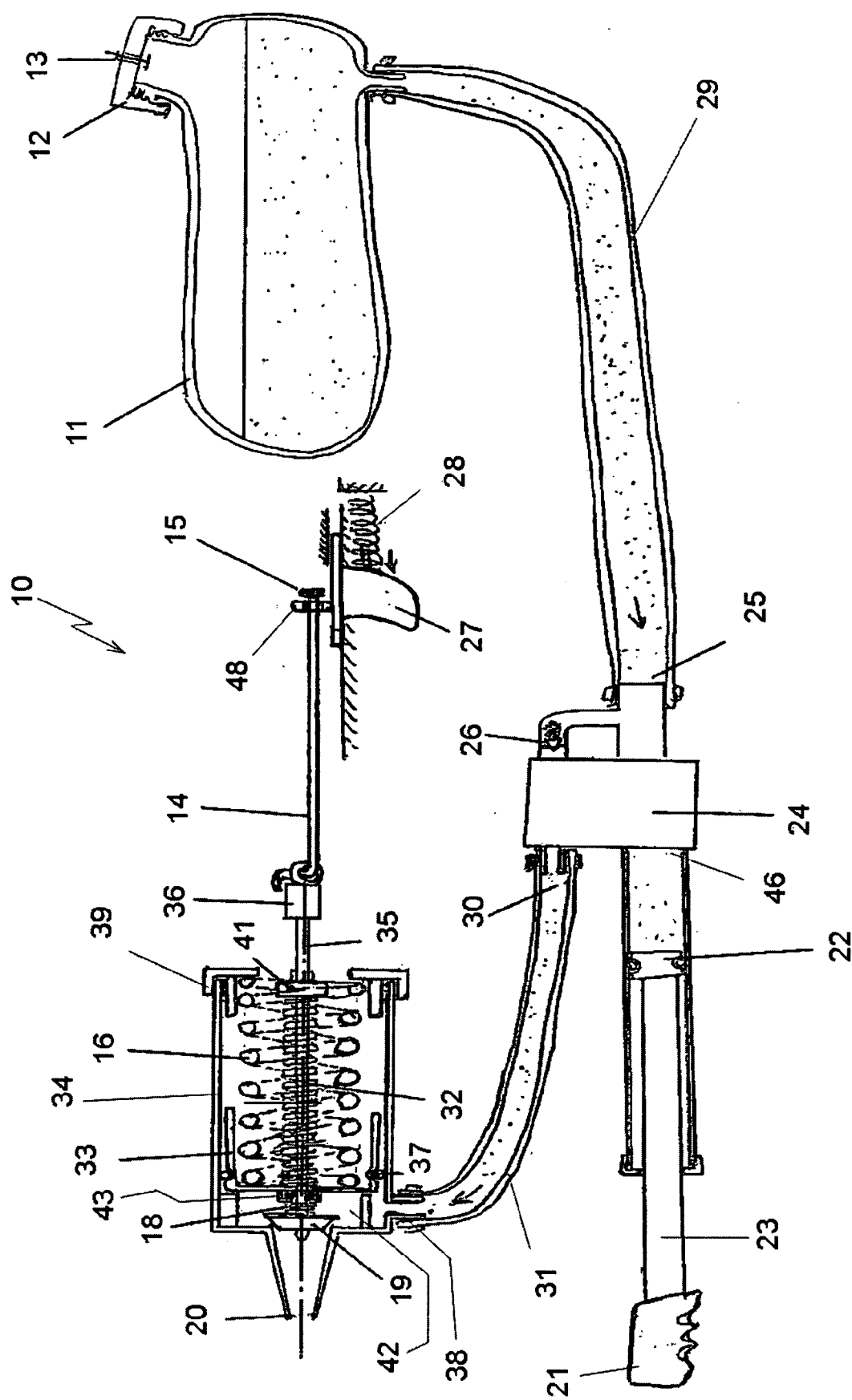
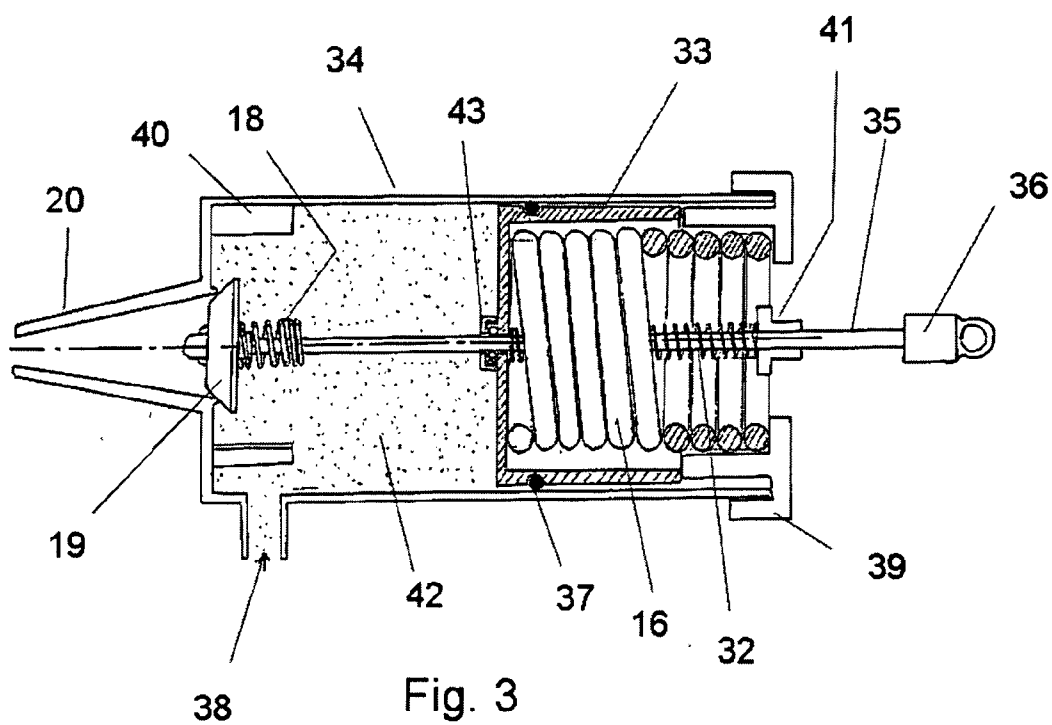
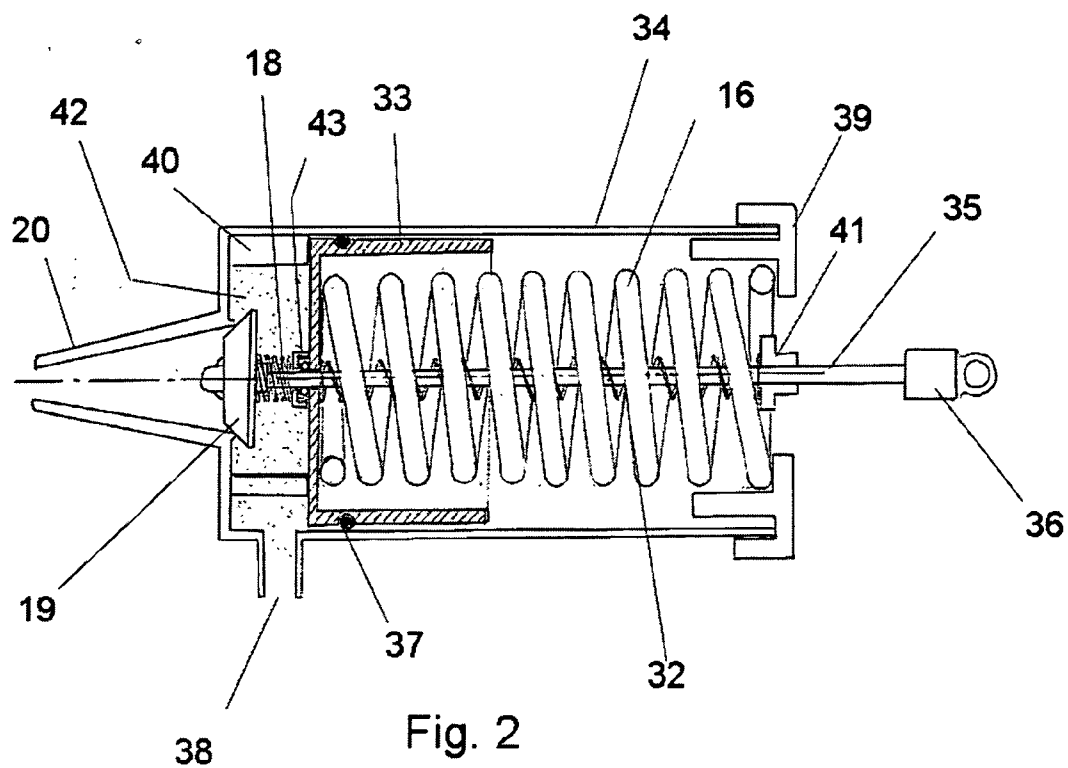
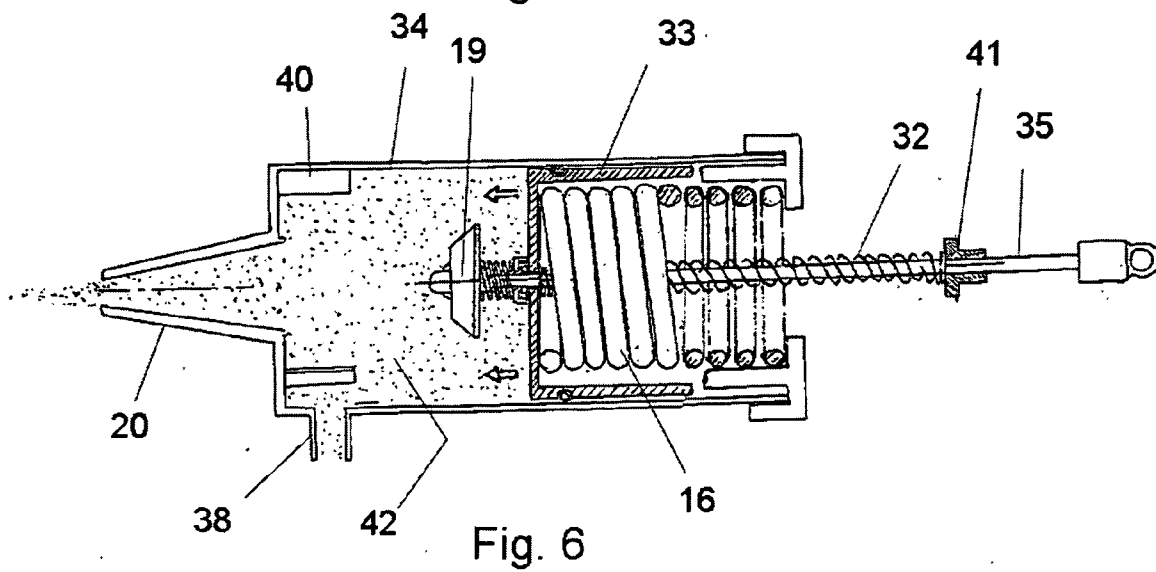
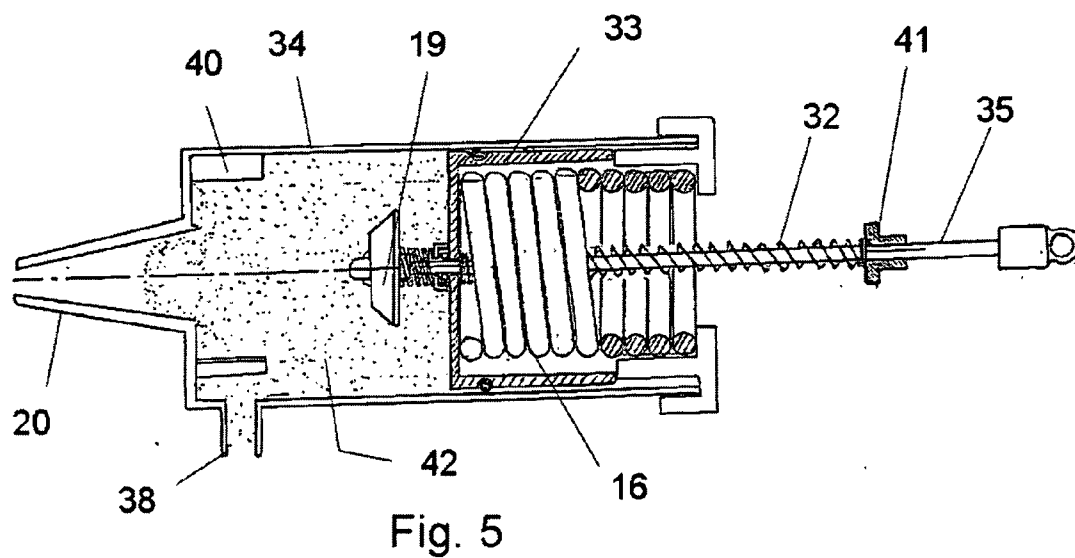
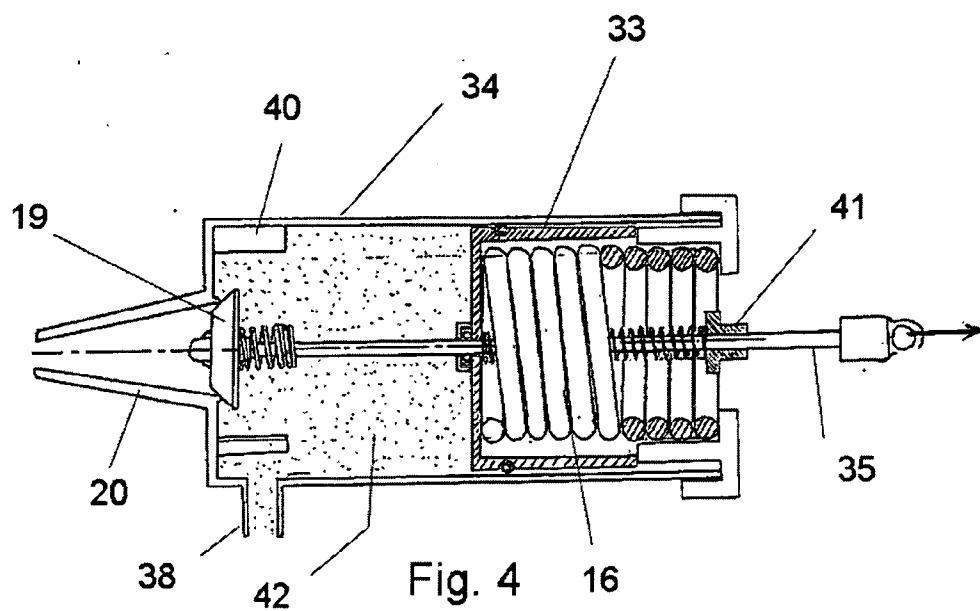


Fig. 1





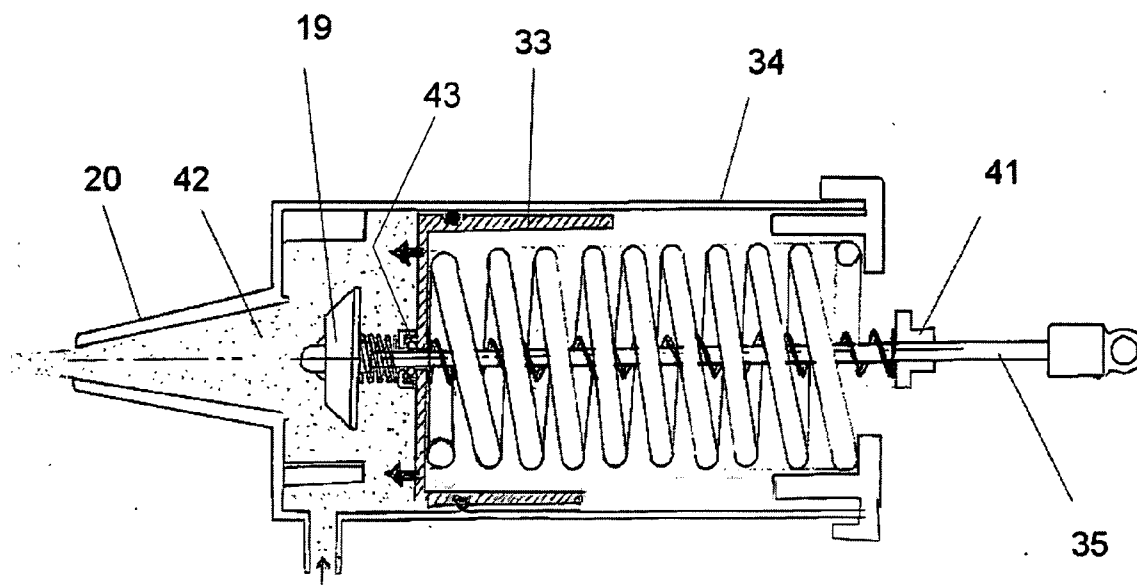


Fig. 7

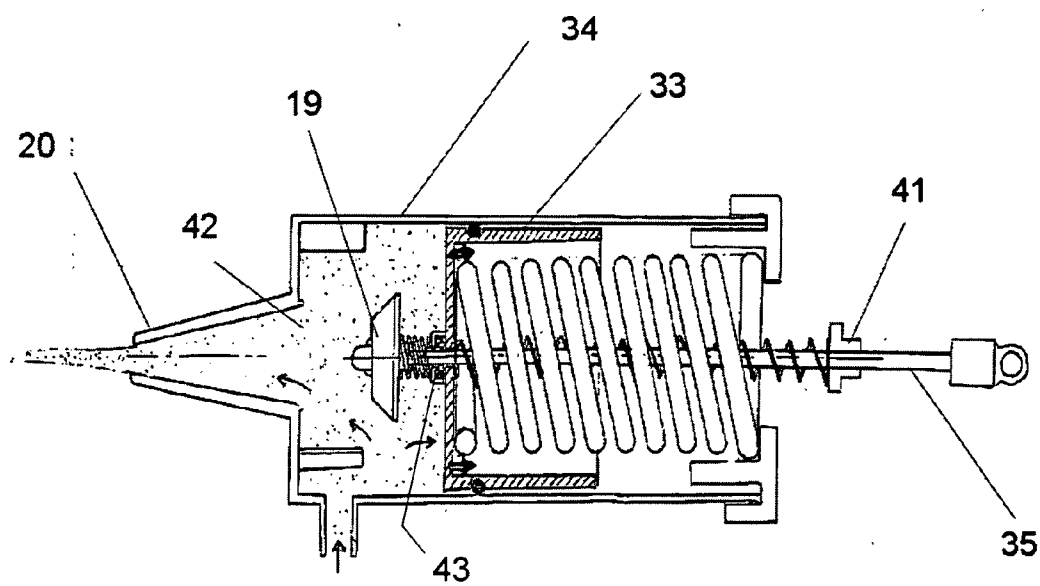


Fig. 8

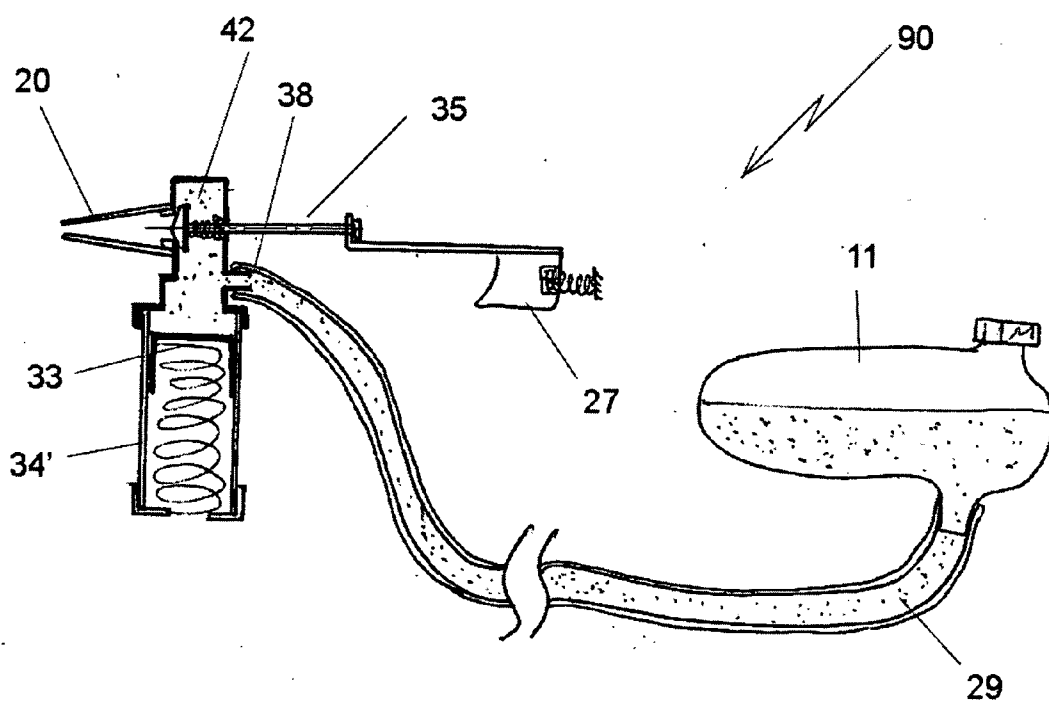


Fig. 9

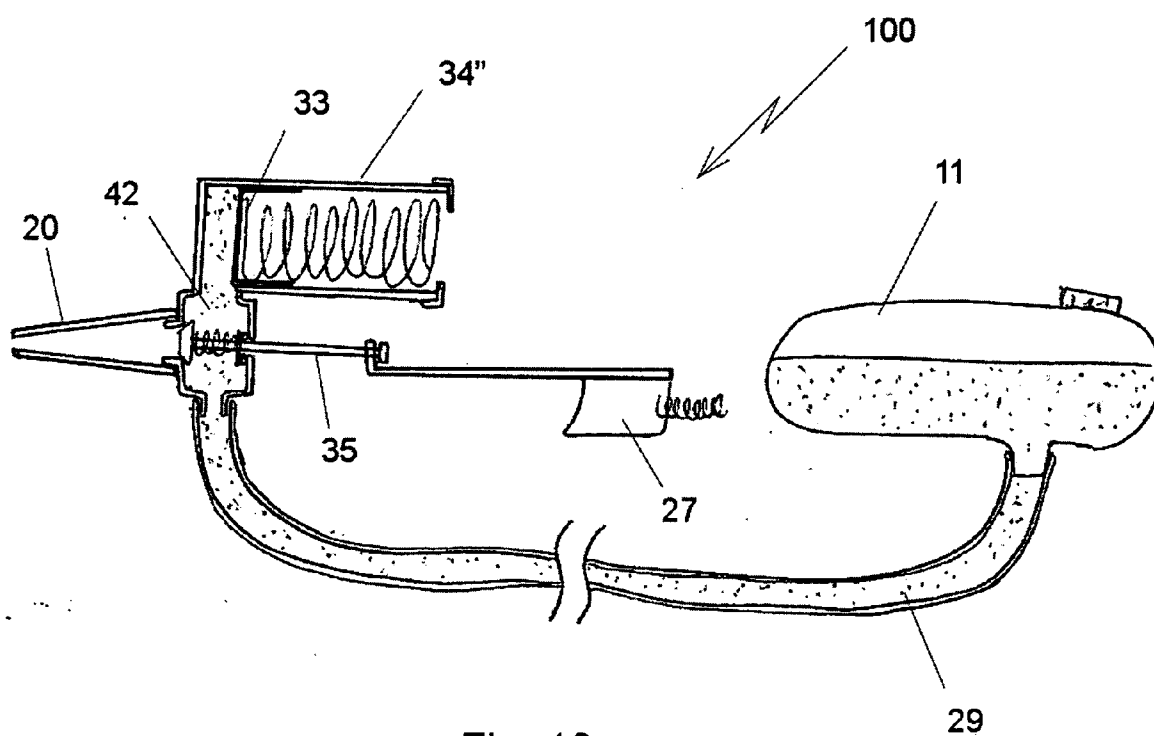


Fig. 10

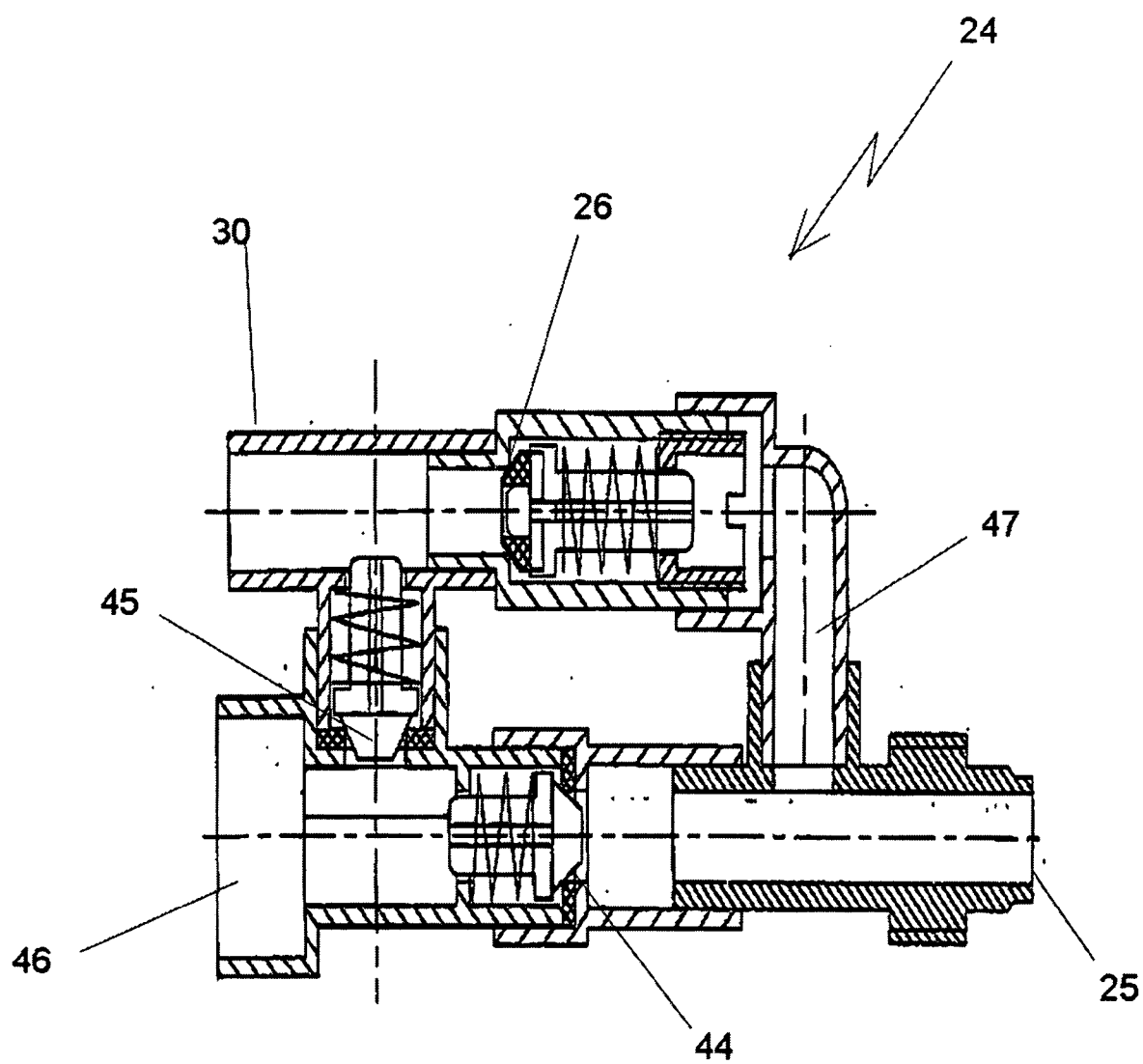


Fig. 11

PRESSURISED TOY WATER GUN

BACKGROUND OF THE INVENTION

[0001] The present invention relates to toy water guns. More particularly although not exclusively, the invention relates to a pressurised toy water gun that provides an instantaneous and continuous water jet.

[0002] Known toy water guns include a hand-activated reciprocating pump mechanism for pumping air into a pressure storage device. It is necessary that the hand pump be activated repeatedly immediately upon commencement of firing. This can delay gameplay and unnecessarily tire the user.

OBJECTS OF THE INVENTION

[0003] It is an object of the present invention to overcome or substantially ameliorate at least one of the above disadvantages and/or more generally to provide an improved toy water gun.

DISCLOSURE OF THE INVENTION

[0004] There is disclosed herein a toy gun comprising:

- [0005] a reservoir for storing liquid;
- [0006] a firing chamber;
- [0007] a nozzle extending from the firing chamber;
- [0008] a valve for closing off the nozzle;
- [0009] a firing piston in communication with the firing chamber;
- [0010] a firing spring biasing the firing piston in a direction to diminish the volume of the firing chamber;
- [0011] a pump receiving liquid from the reservoir and delivering same under pressure to the firing chamber whereupon the firing piston moves against the spring;
- [0012] a trigger connected with the valve so as to open the valve to eject liquid contents of the firing chamber through the nozzle under pressure.

[0013] Preferably, the pump comprises a reciprocating priming piston and an inlet connected via a conduit to the reservoir and an outlet connected via a conduit to the firing chamber.

[0014] Preferably, the valve includes a valve stem extending along an axis of the nozzle.

[0015] Preferably, the firing spring is a coil spring having a longitudinal axis that is common with that of the nozzle.

[0016] Preferably, the valve stem passes through the firing piston.

[0017] Preferably, the toy gun further comprises a valve closing spring surrounding the valve stem and biasing the valve into a closed position.

[0018] Preferably, the toy gun further comprises a valve opening spring surrounding the valve stem behind the firing piston.

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] Preferred forms of the present invention will now be described by way of example with reference to the accompanying drawings, wherein:

[0020] FIG. 1 is a schematic cross-sectional elevation of the major working components of a toy gun,

[0021] FIG. 2 is a schematic cross-sectional elevation of the firing cylinder of the toy gun in a low pressure state,

[0022] FIG. 3 is a schematic cross-sectional elevation of the firing cylinder in an energised state,

[0023] FIG. 4 is a schematic cross-sectional elevation of the firing cylinder at commencement of trigger activation,

[0024] FIG. 5 is a schematic cross-sectional elevation of the firing cylinder immediately after the trigger has been squeezed,

[0025] FIG. 6 is a schematic cross-sectional elevation of the firing cylinder as the firing piston commences forward movement to eject liquid from the firing chamber through the nozzle,

[0026] FIG. 7 is a schematic cross-sectional elevation of the firing cylinder as liquid is being ejected through the nozzle,

[0027] FIG. 8 is a schematic cross-sectional elevation of the firing cylinder with the firing piston in the same position as shown in FIG. 7, but where the priming mechanism is activated continuously during firing to maintain a flow of liquid into the firing chamber and through the nozzle,

[0028] FIG. 9 is a schematic cross-sectional elevation of the major working components of another toy gun having an alternative firing cylinder and firing piston configuration, with the hydraulic diverter omitted from the drawing,

[0029] FIG. 10 is a schematic cross-sectional elevation of yet another toy gun arrangement having a further alternative firing cylinder and firing piston configuration, with the hydraulic diverter omitted from the drawing, and

[0030] FIG. 11 is a schematic cross-sectional elevation showing the internal components of the hydraulic diverter forming part of the toy gun.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0031] In FIGS. 1 to 8 of the accompanying drawings the major components of a toy gun 10 are depicted schematically. The toy gun body is omitted from the drawings as the body can take on any desired shape and configuration.

[0032] The gun 10 includes a water reservoir 11 having a screw cap 12 upon its filling opening. The screw cap 12 has an air vent 13. Extending from a drain point of the reservoir 11 is a conduit 29. This attaches to the inlet 25 of a hydraulic diverter 24. A bypass 47 from the inlet 25 to the hydraulic diverter 24 includes a relief valve 26 as depicted. The internal structure of the hydraulic diverter 24 it depicted in FIG. 11.

[0033] Extending from the body of the hydraulic diverter 24 and attached thereto at attaching flange 46 is an elongate priming cylinder within and from which a reciprocating priming piston 22 extends. A priming piston 22 is mounted at the proximal end of a priming rod 23. The distal end of the priming rod is provided with a handle 21.

[0034] The hydraulic diverter 24 includes an outlet 30 to which another conduit 31 is attached. The other end of the conduit 31 is affixed to an inlet 38 of a firing cylinder 34.

[0035] The firing cylinder 34 has located within it a reciprocating firing piston 33 having a sealing o-ring 37 extending thereabout and in sealing engagement with the inner cylindrical wall of the firing cylinder 34.

[0036] Forward of the firing piston 33 is a firing chamber 42 into which the inlet 38 delivers liquid from the hydraulic diverter 24.

[0037] A valve 19 seals against a valve seat defined by the back edge of the nozzle 20. The valve 19 is fixed upon a valve stem 35 that extends along the axis of the nozzle 20, through the firing piston 33 at which there is provided a sliding seal 43 and preferably through an opening in an end cap 39 at back end of the firing cylinder 34.

[0038] The end cap 39 is threaded or otherwise secured to the firing cylinder 34 and retains a firing compression coil spring 16 against the back of the firing piston 33. The valve stem 35 extends through the centre of the firing spring 16 along its axis. Immediately behind the valve 19 is a short light valve closing spring 18 that serves to press the valve 19 against the valve seat when the firing piston 33 is in the forward position as depicted in FIG. 2. This complements the forward force applied by the hydraulic pressure of any liquid within the firing chamber 42 in maintaining the valve 19 closed.

[0039] A firing piston buffer 40 is provided in the firing chamber 42 to limit the extent of forward movement of the firing piston 33. Also surrounding the valve stem 35 is a long valve opening spring 32. This bears in compression against a stopper 41 that is fixed upon the valve stem 35.

[0040] Mounted at a convenient location upon the toy gun body is a firing trigger 27. The tail end of the link 14 passes through an aperture in a connecting tag 48 of the firing trigger 27. There is a small pulling lug 15 fixed at the tail end of the link 14 against which the back face of the connecting tag 48 bears to draw the link 14 back when the trigger 27 is squeezed in use. This sliding arrangement allows the pulling lug 15 to separate from the connecting tag 48 so that the valve stem can bounce backward (faster/further than the trigger) after the trigger is squeezed. A trigger spring 28 biases the trigger 27 forward and serves to return the trigger to its rest position after it is released.

[0041] In use and prior to pulling the trigger, the pump handle 21 is activated just once or twice so that the hydraulic diverter 24 forces liquid from the reservoir into the firing chamber 42 so that the firing piston 33 moves backwards from the position depicted in FIG. 2 against the force of the firing spring 34 to the position depicted in FIG. 3. During the low-pressure phase, the valve 19 is sealed in part by closing spring 18 against the valve seal at the back of the nozzle 20. The closing spring 18 helps to seal the valve, but once piston 33 moves backward, the closing spring has no effect because it moves away from the front of the piston and any sealing force is provided solely by hydraulic pressure.

[0042] Once the firing piston 33 has moved to its backmost position, any further reciprocation of handle 21 will result in opening of relief valve 26 and the return of over-pumped liquid to the conduit 29. Such bypassing discontinues only when pumping ceases or the trigger 27 is squeezed.

[0043] As shown in FIG. 4, nothing is emitted from the nozzle 20 until the trigger 27 is squeezed to draw the link 14 and valve stem 35 away from nozzle 20 as shown in FIGS. 5 and 6.

[0044] Irrespective of whether the firing chamber is full of liquid or otherwise, the pulling force on the valve stem by the trigger will open the valve 19 immediately. The pulling force to release/open the valve must exceed the combined hydraulic force on the back of the valve minus the spring force applied by valve opening spring 32. Once this threshold is exceeded, the valve 19 will rapidly bounce away from the nozzle 20 by action of the valve opening spring 32. Energy stored in the firing spring 16 then pushes the piston 33 rapidly forward so that liquid is emitted past valve 19 through the nozzle 20. If the player does not pump the handle 21, the jet of water from nozzle 20 ceases when the firing piston 33 has reached the buffer 40 and the valve 19 has returned to its rest/closed position. The pump handle 21 can be repeatedly and continuously reciprocated to draw liquid from the reservoir 11 and

deliver it to the firing chamber for dispatch through the nozzle 20 under pump action of the priming piston.

[0045] It will be understood that the hydraulic diverter 24 has two strokes—namely a pumping stroke and a suction stroke. During the pumping stroke, liquid flows into the firing chamber 42. During the suction stroke, no liquid flows into the firing chamber and as such the firing piston 33 will move forward. By correctly calibrating the nozzle dimensions to the pump throughput, a continuous flow of liquid from the nozzle 20 can be achieved upon reciprocation of handle 21.

[0046] Referring to FIG. 11, the pumping action will further be described with reference to the internal structure of the hydraulic diverter 24. As mentioned earlier, the priming cylinder is attached to the hydraulic diverter at attachment flange 46. Immediately behind the attachment flange 46 is an outlet valve 45. When the priming piston 22 moves toward the hydraulic diverter 24 (the pumping stroke), the outlet valve 45 opens to allow the passage of water through outlet 30. There is an inlet valve 44 closing a passage between the inlet 25 and the attachment flange 46. As the priming piston moves toward the hydraulic diverter 24, the inlet valve 44 is forced closed so that all the liquid between the inlet valve 44 and priming piston 22 passes through the outlet valve 45. When the priming piston 22 is drawn away from the hydraulic diverter 24 (the suction stroke), the outlet valve 45 closes and at the inlet valve 44 opens to allow liquid to be drawn in from reservoir 11.

[0047] When the user ceases reciprocation of the handle 21, the firing piston 33 will returned to the position depicted in FIG. 3 and any further pumping will simply result in a back flow of water through the relief valve 26 as described earlier.

[0048] The embodiments depicted in FIGS. 9 and 10 function in the same manner as described above. These embodiments 90 and 100 illustrate examples in which the valve stem 35 need not extend through the firing piston 33. In FIG. 9, the firing cylinder 34' depends from the firing chamber 42, whereas in FIG. 10, the firing cylinder 34" is positioned above the firing chamber 42. In neither embodiment does the valve stem 35 extended through the firing piston 33. As such, a seal (like 43 in the preceding embodiment) can be omitted.

[0049] It should be appreciated that modifications and alterations obvious to those skilled in the art are not to be considered as beyond the scope of the present invention. For example, rather than providing a linearly reciprocating piston rod and priming piston 22, the hydraulic diverter 24 could be activated by a pivotally reciprocating handle attached to internal rotary pump parts.

1. A toy gun comprising:
 - a reservoir for storing liquid;
 - a firing chamber;
 - a nozzle extending from the firing chamber;
 - a valve for closing off the nozzle;
 - a firing piston in communication with the firing chamber;
 - a firing spring biasing the firing piston in a direction to diminish the volume of the firing chamber;
 - a pump receiving liquid from the reservoir and delivering same under pressure to the firing chamber whereupon the firing piston moves against the firing spring;
 - a trigger connected with the valve so as to open the valve to eject liquid contents of the firing chamber through the nozzle under pressure.

2. The toy gun of claim 1, wherein the pump comprises a reciprocating priming piston and an inlet connected via a conduit to the reservoir and an outlet connected via a conduit to the firing chamber.

3. The toy gun of claim 1, wherein the valve includes a valve stem extending along an axis of the nozzle.

4. The toy gun of claim 3, wherein the firing spring is a coil spring having a longitudinal axis that is common with that of the nozzle.

5. The toy gun of claim 4, wherein the valve stem passes through the firing piston.

6. The toy gun of claim 3, further comprising a valve closing spring surrounding the valve stem and biasing the valve into a closed position.

7. The toy gun of claim 5, further comprising a valve opening spring surrounding the valve stem behind the firing piston.

* * * * *